INCH-POUND

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DETAIL SPECIFICATION
CARTRIDGE, 105MM, ILLUMINATING, M314A3
LOADING, ASSEMBLING AND PACKING

Inactive for new design after 14 March 1997

This specification is approved for use by the U.S. Army Armament Research Development and Engineering Center (ARDEC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers Projectile, 105MM, Illuminating, M314A3 Loading, Assembly and Packing (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in section 3 or 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to the Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, NJ 07806-5000 or emailed to ardecstdzn@conus.army.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2.2 Government Documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENCE SPECIFICATIONS

MIL-DTL-32260	-	Primer, Percussion M28A2

DEPARTMENT OF DEFENCE STANDARDS

MIL-STD-286	-	Propellants, Solid Sampling, Examination and Testing
MIL-STD-1234	-	Pyrotechnics, Sampling, Inspection
MIL-STD-1168	-	Lot Numbering of Ammunition and Ammunition Data Card
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Unless otherwise indicated, copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

7549072	-	Sealed wood packing box
7549073	-	Container, Ammo, Fiber, M105A3, for 105MM Howitzer Ammunition
9201268	-	Procedure, Light Output Measurement
9206744	-	Projectile, 105MM, Illum., M314A3, Loading Assembly
9206821	-	Cartridge, 105MM, Illum., M314A3
12934901	-	Packing and marking for container ammo metal, PA117, for 105MM Howitzer CTG. Packed

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		in PA55 or M105A3
19-48-4231/45-20PM1006	-	Unitization procedures for ammunition components packed in a cylindrical metal or plastic container on a 4-way entry metal pallet

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) Specifications

DTL-8838130	-	Primer, Percussion M28B2

(Copies of these drawings may be requested online at Pica.drawing.request@conus.army.mil or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E 18	-	Standard Test Methods for Rockwell Hardness of Metallic Materials

(Copies of this document are available online at http://www.astm.org or from ASTM International, 100 Barr harbor Drive, West Conshohocken, PA 19428-2959.)

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 100-199

(Copies of this document are available online at http://www.gpoaccess.gov/cfr/index.html or from the Superintendent of Documents, U.S. Government Printing Office, P.O.Box 37082, Washington, D.C. 20013-7082. Orders should specify, "49 CFR 100-199 (latest revision)").

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 First article inspection. When specified (see 6.2) a sample shall be subjected to first article inspection in accordance with 4.2

3.2 Components and assemblies.

3.2.1 Illuminating canister prior to loading illuminating composition. The components and assemblies shall be in accordance with all requirements specified on drawing 9206749.

3.2.2 Illuminating canister. The components and assemblies shall be in accordance with all requirements specified on drawing 9206749.

3.2.3 Illuminating canister and parachute assembly, prior to crimping. The components and assemblies shall be in accordance with all requirements specified on drawing 9206746.

3.2.4 Illuminating canister and parachute assembly. The components and assemblies shall be in accordance with all requirements specified on drawing 9206746.

3.2.5 Projectile loading assy., prior to assy. of illuminating canister & parachute assembly. The components and assemblies shall be in accordance with all requirements specified on drawing 9206744.

3.2.6 Projectile loading assy., prior to securing base plug & closing plug. The components and assemblies shall be in accordance with all requirements specified on drawing 9206744.

3.2.7 Projectile loading assembly. The components and assemblies shall be in accordance with all requirements specified on drawing 9206744.

3.2.8 Primer, prior to assy. to cartridge case. The components and assemblies shall be in accordance with all requirements specified on drawing 9206821.

3.2.9 Cartridge case, prior to assembling to projectile assembly. The components and assemblies shall be in accordance with all requirements specified on drawing 9206821.

3.2.10 Cartridge assembly. The components and assemblies shall be in accordance with all requirements specified on drawing 9206821.

3.2.11 Unsealed fiber container. The components and assemblies shall be in accordance with all requirements specified on drawing 7549073.

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3.2.12 Sealed fiber container. The components and assemblies shall be in accordance with all requirements specified on drawing 7549073.

3.2.13 Sealed wood packing box. The components and assemblies shall be in accordance with all requirements specified on drawing 7549072.

3.2.14 Sealed metal container. The components and assemblies shall be in accordance with all requirements specified on drawing 12934901.

3.2.15 Palletized projectile, M314A3.The components and assemblies shall be in accordance with all requirements specified on drawing 19-48-4231/45-20PM1006.

3.2.16 Alternative, fiberboard liner canister assembly. The components and assemblies shall be in accordance with all requirements specified on drawing 130009273.

3.2.17 Alternative, illuminating canister loading assembly (fiberboard liner). The components and assemblies shall be in accordance with all requirements specified on drawing 13009272.

3.2.18 Alternative, plate baffle. The components and assemblies shall be in accordance with all requirements specified on drawing 130009275.

3.2.19 Alternative fiberboard liner. The components and assemblies shall be in accordance with all requirements specified on drawing 13009274.

3.3 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.4 Moisture content and burning time.

3.4.1 First fire composition. The moisture content of the first fire composition at the loading station, prior to loading shall not exceed 0.30 percent.

3.4.2 Illuminant composition. The moisture content of the illuminant composition at the loading station prior to loading shall not exceed 0.30 percent. Composition must be pressed within 8 hours of mixing.

3.4.3 Luminosity and burning time. The average luminosity of the canister assemblies shall not be less than 525,000 candlepower. The burning time of the canister assemblies shall not be less than 60 seconds.

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3.4.4 Fiber board liner moisture content. The moisture content of the fiber board liner shall be 7.0% max during production.

3.5 Proving ground. The cartridge shall comply with the following requirements.

3.5.1 Functioning. The cartridge shall function satisfactorily (see 6.9) and shall provide effective illumination (see 6.10) for 55.0 seconds.

3.5.2 Projectile safety. There shall be no premature burst or indication of metal parts separation in the gun or in the first 100 yards of flight.

3.6 Workmanship. The requirements for workmanship are as follows:

3.6.1 Threads. Threads shall be undamaged for the entire minimum length or depth.

3.6.2 Burr and sharp edges. No part shall have a burr or sharp edge which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.6.3 Foreign matter. No part or assembly shall contain dirt, grease, chips, oil, rust, corrosion or other foreign matter.

3.6.4 Painting and marking. Painting and marking shall comply with the requirements of drawing 12934901. All paint shall be dry to the touch before packing for shipment. The paint shall be considered dry when the soft tacky condition no longer exists and the paint feels firm when touched with a finger using moderate pressure.

3.6.5 Protection of rotating band. The machined rotating band shall be protected from being nicked, burred or otherwise damaged in subsequent handling prior to shipment.

3.6.6 Protective coatings. Protective coatings shall be in accordance with the applicable drawing and the specifications listed thereon.

3.6.7 Cloth. All cloth shall be free of tears, rips and foreign matter. The stitches in the cloth shall be neat and continuous without breaks, tears or foreign matter.

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4. VERIFICATION

TABLE I. Requirement/verification cross reference matrix

Methods of Verification N/A – Not Applicable 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test	Classes of Verification A– First Article B– Acceptance

Section 3 Requirements	Description	Verification Method					Verification Class		Section 4 Verification
		N/A	1	2	3	4	A	B	
3.1	First Article				X	X	X	X	4.2
3.2	Components and assemblies								
3.2.1	Illuminating canister prior to loading illuminating composition				X	X	X	X	4.3.2.1
3.2.2	Illuminating canister				X	X	X	X	4.3.2.2
3.2.3	Illuminating canister & parachute assembly, prior to crimping				X	X	X	X	4.3.2.3
3.2.4	Illuminating canister & parachute assembly				X	X	X	X	4.3.2.4
3.2.5	Projectile loading assembly, prior to assy. of illuminating canister & parachute assy.				X	X	X	X	4.3.2.5
3.2.6	Projectile loading assy. prior to securing base plug & closing plug				X	X	X	X	4.3.2.6
3.2.7	Projectile loading assy.				X	X	X	X	4.3.2.7
3.2.8	Primer, prior to assy. to cartridge case				X	X	X	X	4.3.2.8
3.2.9	Cartridge case prior to assy. to projectile assy.				X	X	X	X	4.3.2.9
3.2.10	Cartridge assembly				X	X	X	X	4.3.2.10
3.2.11	Unsealed fiber container				X	X	X	X	4.3.2.11
3.2.12	Sealed fiber container				X	X	X	X	4.3.2.12
3.2.13	Sealed wood packing box				X	X	X	X	4.3.2.13
3.2.14	Sealed metal container				X	X	X	X	4.3.2.14
3.2.15	Palletized projectile				X	X	X	X	4.3.2.15
3.2.16	Alternate, fiberboard liner canister assembly				X	X	X	X	4.3.2.16
3.2.17	Alternate, illuminating canister loading assembly				X	X	X	X	4.3.2.17
3.2.18	Alternate, plate, baffle				X	X	X	X	4.3.2.18
3.2.19	Alternate, fiberboard liner				X	X	X	X	4.3.2.19
3.3	Ammunition lot numbering				X	X	X	X	4.3.1
3.4	Moisture content and burning time					X	X	X	
3.4.1	First fire composition					X	X	X	4.4.2.1
3.4.2	Illuminant composition					X	X	X	4.4.2.2
3.4.3	Luminosity and burning time				X	X	X	X	4.4.3
3.4.4	Fiber board liner moisture content					X	X	X	4.4.2.4

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TABLE I. Requirement/verification cross reference matrix - Continued

Methods of Verification N/A – Not Applicable 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test	Classes of Verification A- First Article B- Acceptance

Section 3 Requirements	Description	Verification Method					Verification Class		Section 4 Verification
		N/A	1	2	3	4	A	B	
3.5	Proving ground								
3.5.1	Functioning				X	X	X	X	4.4.1.1
3.5.2	Premature burst				X	X	X	X	4..4.1.2
3.5.3	Safety				X	X	X	X	4.4.1.3
3.6	Workmanship								
3.6.1	Threads				X		X	X	4.3.2.1 – 4.3.2.19
3.6.2	Burr and sharp edges				X		X	X	4.3.2.1 – 4.3.2.19
3.6.3	Foreign matter				X		X	X	4.3.2.1 – 4.3.2.19
3.6.4	Painting and marking				X		X	X	4.3.2.1 – 4.3.2.19
3.6.5	Protection of rotating band				X		X	X	4.3.2.1 – 4.3.2.19
3.6.6	Protective coatings				X		X	X	4.3.2.1 – 4.3.2.19
3.6.7	Cloth				X		X	X	4.3.2.1 – 4.3.2.19

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4.1 Classification of inspection. The verification requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. When specified, a sample shall be subjected to first article verification in accordance with Table II.

4.2.1 First article quantity. First article verification shall be performed on the quantity of items as indicated in Table II.

4.2.2 Inspections to be performed. The first article verification shall be performed in accordance with Table II.

4.2.3 First article rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

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TABLE II - First article inspection

PARAGRAPH	TITLE Projectile, 105MM, illuminating, M314A3	SHEET 1 OF 4	DRAWING NUMBER NEXT HIGHER ASSEMBLY	
CLASSIFICATION	EXAMINATION OR TEST	number of samples	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Illuminating canister (Dwg. 9206748) (Prior to loading illuminating composition) Examination for defects	5	3.2.1	4.3.2.1
	Illuminating canister (Dwg. 9206749) Illuminant composition test Luminosity and burning time Examination for defects	4 oz 5 pair 5	3.4.2 3.4.3 3.2.2	4.4.2.2 4.4.3 4.3.2.2
	First fire composition First fire composition test	4 oz	3.4.1	4.4.2.1
	Illuminating canister & parachute assy (Dwg. 9206746) (Prior to crimping) Examination for defects	5	3.2.3	4.3.2.3
	Illumination canister & parachute assy (Dwg. 9206746) Examination for defects	5	3.2.4	4.3.2.4

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TABLE II - First article inspection - Continued

PARAGRAPH	TITLE Projectile, 105MM, illuminating, M314A3	SHEET 2 OF 4	DRAWING NUMBER NEXT HIGHER ASSEMBLY	
CLASSIFICATION	EXAMINATION OR TEST	number of samples	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Projectile loading assy. (Dwg. 9206744) (Prior to assy. of illuminating canister & parachute assy.) Examination for defects	5	3.2.5	4.3.2.5
	Projectile loading assy. (Dwg. 9206744) (Prior to securing base plug and closing plug) Examination for defects	5	3.2.6	4.3.2.6
	Projectile loading assy. (Dwg. 9206744) Examination for defects Proving Ground Tests (Note 1)	5 80	3.2.7 3.5	4.3.2.7 4.4.1
	Primer (Dwg. 9206821) (prior to assy. to cartridge case) Examination for defects	5	3.2.8	4.3.2.8
	Cartridge case (Dwg. 9206821) (Prior to assembling projectile assy.) Examination for defects	5	3.2.9	4.3.2.9

Note 1. Ballistic testing will be performed subsequent to successful completion of all dimensional and visual inspections outlined in Table II.

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TABLE II - First article inspection - Continued

PARAGRAPH	TITLE Projectile, 105MM, illuminating, M314A3	SHEET 3 OF 4	DRAWING NUMBER NEXT HIGHER ASSEMBLY	
CLASSIFICATION	EXAMINATION OR TEST	number of samples	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Cartridge assembly (Dwg. 9206821) Examination for defects	5	3.2.10	4.3.2.10
	Container, ammo. fiber, M105A3, for 105MM howitzer ammunition (Dwg. 7549073) Examination for defects	5	3.2.12	4.3.2.11/4.3.2.12
	Packing and marking for container ammo. metal, PA117, for 105MM howitzer ctg. packed in PA55 or M105A3 (Dwg. 12934901) Examination for defects	5	3.2.14	4.3.2.14
	Palletized projectile, M314A3: packed in accordance with drawing 19-48-4231/45-20PM1006 Examination for defects	1 pallet	3.2.15	4.3.2.15

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PARAGRAPH	TITLE Projectile, 105MM, illuminating, M314A3	SHEET 4 OF 4	DRAWING NUMBER NEXT HIGHER ASSEMBLY	
CLASSIFICATION	EXAMINATION OR TEST	number of samples	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Alternative, fiberboard liner canister assembly prior to loading illuminating composition (Dwg. 130009273) Examination for defects	5	3.2.16	4.3.2.16
	Alternative, illuminating canister loading assembly (fiberboard liner) (Dwg. 13009272) Examination for defects	5	3.2.17	4.3.2.17
	Alternative, plate, baffle (Dwg. 13009275) Examination for defects	10	3.2.18	4.3.2.18
	Alternative, fiberboard liner (Dwg. 13009274) Examination for defects	10	3.2.19	4.3.2.19

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4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-STD-1916, Acceptance by Tables, paragraph 4.2. In addition, inspection lots of projectile shall contain:

a. Projectile metal parts assemblies from one lot interfix from one manufactures
b. Expelling charge assemblies from one lot interfix number from one manufacturer
c. Canister assemblies from one lot interfix number from one manufacturer
d. Parachute assemblies from one lot interfix number from one manufacturer
e. Primers of one lot interfix number from one manufacturer
f. Cartridge assemblies of one lot interfix number from one manufacturer

4.3.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for critical, major and minor characteristics are defined in MIL-STD-1916. Unless otherwise specified, inspection level IV shall be used for all characteristics defined as majors and inspection level II for all minor characteristics. Critical characteristics shall be addresses in accordance with MIL-STD-1916.

b. Conformance inspection. Conformation inspection shall be performed accordance with paragraph 4.3.2.1 through 4.3.2.19. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests.

c. Alternative inspections. Alternative conformance inspections shall be submitted and approved in accordance with MIL-STD-1916

d. Tool control. Dimensions marked tool control shall be gaged at the beginning of production and whenever tooling is replaced. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided there is an established correlation between the tool dimensions and the component dimensions prior to the start of production. There shall be adequate surveillance to assure that the drawing requirements are being met throughout the life of the tool.

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.1	TITLE Illuminating canister, prior to loading illuminating composition	SHEET 1 OF 1	DRAWING NUMBER 9206749
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Any component missing, damaged or improperly assembled	Level IV	3.2.1	Visual
102	Liner composition present on open end of canister between brakes	Level IV	3.2.1	Visual
Minor				
201	Coating missing or inadequate	Level II	3.2.1	Visual
202	Point stake missing or inadequate	Level II	3.6	Visual
203	Evidence of poor workmanship (other than coating)	Level II	3.6	Visual/Manual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.2	TITLE Illuminating canister	SHEET 1 OF 1	DRAWING NUMBER 9206749
			NEXT HIGHER ASSEMBLY 9206746

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Total length, maximum (max.)	Level IV	3.2.2	AIE
102	Any component missing, damaged or improperly assembled	Level IV	3.2.2	Visual
103	Disc assembly above flush	Level IV	3.2.2	Visual
104	First fire composition	4.4.2.1	3.4.1	4.4.2.1
105	Illuminating composition test	4.4.2.2	3.4.2	4.4.2.2
106	Static functioning	4.4.3	3.4.3	4.4.3
Minor				
201	Disc assembly coating missing or incomplete (prior to assy. into canister)	Level II	3.6	Visual
202	Evidence of poor workmanship	Level II	3.6	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.3	TITLE Illuminating canister and parachute assembly, prior to crimping	SHEET 1 OF 1	DRAWING NUMBER 9206746 / NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Length of center wire	Level IV	3.2.3	AIE
Minor				
201	Staking missing or improper assy. of swivel nut	Level II	3.2.3	Visual
202	Wire not frayed or improperly soldered	Level II	3.2.3	Visual
203	Evidence of poor workmanship	Level II	3.6	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.4	**TITLE** Illuminating canister and parachute assembly	**SHEET** 1 **OF** 1	**DRAWING NUMBER** 9206746
			NEXT HIGHER ASSEMBLY 9206744

CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical	None defined			
Major				
101	Any component or operation missing or improper	Level IV	3.2.4	Visual
102	Swivel fails to operate freely	Level IV	3.2.4	Visual/Manual
103	Staking or pettman cement missing or improper	Level IV	3.2.4	Visual
104	Parachute torn	Level IV	3.2.4	Visual
Minor				
201	Evidence of poor workmanship	Level IV	3.6	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.5	**TITLE** Projectile loading assy., prior to assy. of illuminating canister & parachute assy.		**SHEET** 1 **OF** 1	**DRAWING NUMBER** 9206744
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical	None defined			
Major				
101	Any component or operation missing or improper	Level IV	3.2.5	Visual
102	Parachute improperly folded (see Dwg. 9206746, sheet 2)	Level IV	3.2.5	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.6	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.6	**TITLE** Projectile loading assy., prior to securing base plug & closing plug	**SHEET** 1 **OF** 1	**DRAWING NUMBER** 9206744
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical	None defined			
Major				
101	Any component missing or improper	Level IV	3.2.6	Visual
102	Parachute wrapper torn	Level IV	3.2.6	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.6	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.7	TITLE Projectile loading assembly	SHEET 1 OF 2	DRAWING NUMBER 9206744 NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Position of shear pin	Level IV	3.2.7	AIE
102	Diameter of bourrelet and rear or rotating band, maximum (if retouched)	Level IV	3.2.7	AIE
103	Gap between base plug & base, max.	Level IV	3.2.7	AIE
104	Total length	Level IV	3.2.7	AIE
105	Shear not staked in place or missing	Level IV	3.2.7	Visual
106	Shear pin(s) above flush	Level IV	3.2.7	AIE
107	Plug disassembly torque, max (if applicable	Level IV	3.2.7	AIE

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.7	TITLE Projectile loading assembly	SHEET 2 OF 2	DRAWING NUMBER 9206744
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201	Color of band incorrect	Level II	3.2.7	Visual
201	Protective coating with bare spot in excess of ¼ square inch	Level II	3.6	Visual
203	Marking missing, misleading or identifiable	Level II	3.6	Visual
204	Evidence of poor workmanship	Level II	3.6	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.8	**TITLE** Primer, prior to assy. to cartridge case		**SHEET** 1 **OF** 1	**DRAWING NUMBER** 9206821
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical	None Defined			
Major	None Defined			
Minor 201	Evidence of poor workmanship	Level II	3.6	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.9	**TITLE** Cartridge case, prior to assembling to projectile assembly		**SHEET** 1 **OF** 1	**DRAWING NUMBER** 9206821
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical	None defined			
Major				
101	Propelling charge bag missing or torn	Level IV	3.2.9	Visual
102	Foil side of charge No. 5 not facing primer	Level IV	3.2.9	Visual
103	Propelling charge assembled incorrectly	Level IV	3.2.9	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.6	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.10	TITLE Cartridge assembly	SHEET 1 OF 2	DRAWING NUMBER 9206821
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Primer above flush	100%	3.2.10	AIE
2	Ballistic test failure(projectile safety)	4.4.1	3.5.2	4.4.1
Major				
101	Fuze fails to withstand the required disassembly torque	Level IV	3.2.10	AIE/Torque test
102	Primer loose or missing	Level IV	3.2.10	Visual/Manual
103	Fuze setting incorrect	Level IV	3.2.10	Visual
104	Profile and diameter of mouth	100%	3.2.10	AIE/4.4.6
105	Ballistic test failure (functioning)	4.4.1.1	3.5.1	4.4.1.1
Minor				
201	Gap between fuze & projectile	Level II	3.2.10	Visual
202	Fuze not staked	Level II	3.2.10	Visual
203	Rotating band damaged	Level II	3.2.10	Visual
204	Marking missing, misleading or unidentifiable	Level II	3.2.10	Visual
205	Evidence of poor workmanship	Level II	3.6	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.11	**TITLE** Unsealed fiber container		**SHEET** 1 **OF** 1	**DRAWING NUMBER** 7549073
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical	None defined			
Major				
101	Packing component missing, damaged or improperly assembled	Level IV	3.2.11	Visual
102	Cartridge cannot be inserted or removed by hand	Level IV	3.2.11	Manual
103	Container covers cannot be removed by hand	Level IV	3.2.11	Manual
104	Insufficient fillers for tight pack	Level IV	3.2.11	AIE
Minor				
201	Evidence of poor workmanship	Level II	3.6	Visual/Manual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.12	TITLE Sealed fiber container	SHEET 1 OF 1	DRAWING NUMBER 7549073 NEXT HIGHER ASSEMBLY 12934901

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Sealing strip incomplete	Level IV	3.2.12	Visual
102	Sealing strip wrinkled	Level IV	3.2.12	Visual
103	Sealing strip material improper	Level IV	3.2.12	Visual/CERT
104	Sealing strip damaged	Level IV	3.2.12	Visual
105	Interior fiber moisture content excessive at time of pack	Level IV	3.2.12	AIE
106	Component missing	Level IV	3.2.12	Visual
107	Marking misleading, missing, incorrect, or illegible	Level IV	3.2.12	Visual
108	Tight pack	Level IV	3.2.12	Manual/Audible
Minor				
201	Tear tab length below minimum	Level II	3.2.12	AIE
202	Component damages	Level II	3.2.12	Visual
203	Evidence of poor workmanship	Level II	3.6	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.13	TITLE Sealed wood packing box		SHEET 1 OF 1	DRAWING NUMBER 7549072
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical	None defined			
Major				
101	Box damaged exposing contents	Level IV	3.2.13	Visual
102	Hardware or strapping missing, broken or loose	Level IV	3.2.13	Visual
103	Board broken	Level IV	3.2.13	Visual
Minor				
201	Strapping or hardware improperly assembled	Level II	3.2.13	Visual/Manual
202	Handle missing or insecure	Level II	3.2.13	Visual/Manual
203	Car seal missing, unsealed or improperly positioned	Level II	3.2.13	Visual
204	Marking missing, misleading or unidentifiable	Level II	3.2.13	Visual
205	Evidence of poor workmanship	Level II	3.6	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.14	TITLE Sealed metal container	SHEET 1 OF 1	DRAWING NUMBER 12934901
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Container damaged exposing contents	Level IV	3.2.14	Visual
102	Metallic seal missing or improperly secured	Level IV	3.2.14	Visual/Manual
103	DoD symbol missing or incorrect	Level IV	3.2.14	Visual
104	Air leak test	100%	3.2.14	4.4.4
105	Screw not properly replaced and torqued after test	Level IV	3.2.14	AIE/Visual
106	Contents loose inside metal container	Level IV	3.2.14	Manual/Audible
107	Interior packing material moisture content excessive at time of pack	Level IV	3.2.14	AIE
108	Cloth strap incorrect width or length	Level IV	3.2.14	AIE
109	Cloth strap assembled incorrectly	Level IV	3.2.14	Visual
110	Cloth strap missing	Level IV	3.2.14	Visual
111	Marking incorrect, missing, or unidentifiable	Level IV	3.2.14	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.6	Manual/Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.15	TITLE Palletized projectile, M314A3; packed in accordance with drawing 19-48-4231/45-20PM1006	SHEET 1 OF 1	DRAWING NUMBER 7549072 / NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Strapping missing	Note 1	3.2.15	Visual
102	Strapping loose	Note 1	3.2.15	Visual/Manual
103	Incorrect number of containers on pallet (except for light load)	Note 1	3.2.15	Visual
104	Containers not all positioned in the same direction	Level IV (Note 2)	3.2.15	Visual
105	Containers not properly interlocked	Level IV (Note 2)	3.2.15	Visual
106	Marking missing, incorrect, or illegible	Level IV (Note 2)	3.2.15	Visual
Minor				
201	Evidence of poor workmanship	Level II (Note 2)	3.6	Visual/Manual

NOTE: 1. Sample size to be the smallest number of pallets that contain a quantity of projectiles that reflect a level IV.
2. If inspection requires pallets to be opened, the entire quantity of projectiles on that pallet may, at the Government inspectors option, be applied to the sample quantity, provided that the pallet is selected randomly

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.16	**TITLE** Alternative, fiberboard liner canister assembly	**SHEET** 1 **OF** 1	**DRAWING NUMBER** 13009273
			NEXT HIGHER ASSEMBLY 13009272

CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical	None defined			
Major				
101	Location of stakes	Level IV	3.2.16	AIE
102	Depth of stake	Level IV	3.2.16	AIE
103	Stake missing or improper	100%	3.2.16	Visual
104	Epoxy missing or incomplete coverage of inside surface of the fiberboard liner	100%	3.2.16	Visual
Minor				
201	Evidence of poor workmanship	100%	3.6	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.17	**TITLE** Alternative, illuminating canister loading assembly (fiberboard liner)	**SHEET** 1 **OF** 2	**DRAWING NUMBER** 13009272
			NEXT HIGHER ASSEMBLY 9206746

CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical	None defined			
Major				
101	First fire composition moisture content	4.4.2.1	3.4.1	4.4.2.1
102	Illuminant composition moisture content	4.4.2.2	3.4.2	4.4.2.2
103	Ignition composition missing or inadequate	100%	3.2.17	Visual
104	Evidence of delamination (after pressing)	100%	3.2.17	Visual
105	Slurry missing or inadequate	100%	3.2.17	Visual
106	Ignition disc assembly missing	100%	3.2.17	Visual
107	Filler ring not flush to 1/16 above flush	100%	3.2.17	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.18	TITLE Alternative, plate, baffle	SHEET 1 OF 1	DRAWING NUMBER 13009275 NEXT HIGHER ASSEMBLY 9206744

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Diameter, max	Level IV	3.2.18	AIE
102	Number of holes	Level IV	3.2.18	Visual
103	Size of holes	Level IV	3.2.18	AIE
Minor				
201	Evidence of poor workmanship	Level II	3.6	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.19	**TITLE** Alternative, fiberboard liner		**SHEET** 1 **OF** 1	**DRAWING NUMBER** 13009274
				NEXT HIGHER ASSEMBLY 13009272
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical	None defined			
Major				
101	Outside diameter	Level IV	3.2.19	AIE
102	Overall length	Level IV	3.2.19	AIE
103	Thickness	Level IV	3.2.19	AIE
104	Moisture Content	Level IV	3.4.4	AIE
Minor				
201	Evidence of poor workmanship	Level II	3.6	Visual

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4.4 Methods of inspection

4.4.1 Proving ground. These tests shall be conducted at a Government proving ground. The sample cartridges shall be fired with M67 propelling charges in a gun for which the ammunition is standard and observation made for conformance to all requirements (see 6.9 and 6.10).

4.4.1.1 Functioning. The sample cartridges shall be fired at zone 7. If any of the sample cartridges used in any of the tests fail to comply with the requirements as specified in paragraph 3.5.2 the lots shall be rejected.

4.4.1.1.1 Production lot sampling. Beginning with the first lot produced and continuing until three consecutive lots have complied with the requirements as specified in paragraph 3.5.1 herein, 80 cartridges shall be selected for this test. If nine or more cartridges fail to function satisfactorily or to provide effective illumination for at least 55 seconds, the lot shall be rejected.

4.4.1.1.2 Production lot reduced sampling. After three consecutive lots have met the criteria of 4.4.1.1.1, 34 cartridges shall be selected for this test. If five or more fail to function satisfactorily or to provide effective illumination for 55 seconds the lot shall be rejected. In the event that three successive lots fail to meet the requirements of 4.4.1.1.2, the sampling provisions of 4.4.1.1.1 shall again apply. When three consecutive lots have again passed the requirements contained therein, the provisions of 4.4.1.1.2 shall be reinstated.

4.4.1.2 Ballistic safety test. The sample cartridges shall be fired with propelling charges adjusted to give 112 percent of the rated maximum pressure for the cartridge. In addition, for the first three production lots from each manufacturer (loading facility); a smear camera shall be used to observe the projectile in flight in order to record any possible malfunctioning.

4.4.1.2.1 Production lot sampling. Beginning with the first lot produced and continuing until three consecutive lots have complied with the requirements as specified in paragraph 3.5.3 herein, 10 cartridges shall be selected for this test. If any cartridge fails to comply with 3.5.2 the lot shall be rejected.

4.4.1.2.2 Production lot reduced sampling. After three consecutive lots have met the criteria of 4.4.1.3.1, three cartridges shall be selected from each lot for this test. If any cartridge fails to comply with the applicable requirements the lot shall be rejected. In the event a lot fails to meet the requirements of 4.4.1.3.2, the sampling provisions of 4.4.1.3.1 shall again apply. When three

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consecutive lots have again successfully passed the requirements contained therein, the provisions of 4.4.1.3.2 shall be reinstated.

4.4.1.3 Refiring. If for any reason the test director considers that the test conditions have detrimentally affected the test results, the test shall be considered invalid and a new test shall be performed with additional rounds as required.

4.4.2 Moisture content.

4.4.2.1 First fire composition. The moisture content of the first fire composition shall be determined in accordance with Standard MIL-STD-1234, Method 101.3. The special solvent to be used shall be equal parts by volume of dimethylformamide and anhydrous methanol.

4.4.2.1.1 First fire composition sampling plan. The contractor shall provide adequate controls to insure compliance with the requirements and shall select a three to four ounce first fire composition sample from each eight hours of production and place it in an airtight labeled container indicating the name of the material and date of production. If the sample, after analysis, fails to comply with the applicable requirement; the eight hours of production represented by the sample shall be rejected.

4.4.2.2 Illuminant composition. The moisture content of the illuminant composition shall be determined in accordance with the Karl Fischer Method as stated in Standard MIL-STD-1234, Method 101.2 up to paragraph 5.3.0. A 50 ml. sample of 1.1 ml dimethylformamidemethyl alcohol solution shall be added into a 250 ml. Erlenmeyer flash (fitted with a rubber stopper) containing 10 grams of the pyrotechnic sample (exercise care to exclude atmospheric moisture). The flask shall be stoppered and shaken vigorously to thoroughly mix solution and pyrotechnic sample. Allow the solution to set 5 - 15 minutes. A blank sample of the 1: 1 dimethylformamide-methanol solution shall be prepared as described above. A 60 ml. portion of methanol shall be transferred through the sample porthole into dry titration beaker containing a plastic coated magnet. Stopper at once. The magnetic stirrer shall be started and the water back-titrated in this solvent by adding a slight excess of Karl for at least half a minute. The solution shall be titrated slowly with the water-in-methanol solution until the exact preliminary and point is reached. All succeeding operations shall be completed within 3.5-0.25 minutes. Twenty-five (25) ml. of supernatant solution shall be rapidly introduced, accurately pipetted from the supernatant liquid into the beaker stirred for thirty seconds by means of the magnetic stirrer. A slight excess (10-15 m l s) of the Karl Fischer reagent shall be added and carefully back-titrated with the standard

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water-in-methanol solution to the final and exact end point. A water determination of the dimethylformamide methanol blank solution shall be conducted as specified above for the extraction of water from the sample. The percent water in the pyrotechnic sample shall be calculated as follows:

Calculation:

AB/ {5(10)}	=	Percent H_2O
Less		Std. Blank
Leaves		Percent Moisture in Sample

Where:

A = Ml. Of Karl Fisher to titrate sample
B = Standardization Factor (mg. /ml.)
10= Constant Factor
5 = Standard Sample Weight

4.4.2.2.1 Illuminant composition sampling plan. The contractor shall provide adequate controls to insure compliance with the requirements and shall select a three to four ounce illuminant composition sample from each eight hours of production and place it in an airtight labeled container indicating the name of the material and date of production. If the sample, after analysis, fails to comply with the applicable requirement; the eight hours of production represented by the sample shall be rejected.

4.4.2.3 Alternate method for first fire and illuminant composition moisture. The moisture content of the first fire composition and illuminant composition shall be determined in accordance with Method T101.4 Standard MIL-STD-286. The temperature setting shall be 100 degrees C and time setting 30 minutes; the sample size shall be between 0.40 and 0.60 grams.

4.4.2.4 Fiberboard liner moisture content. The moisture content of the fiber board liner shall be verified during production (see 6.2).

4.4.3 Luminosity and burning time. The discs shall be removed from the candle prior to testing. In calculating the candle power of any one illuminant assembly, the best continuous 55 seconds of illuminometer readings are to be used. The test shall be conducted utilizing equipment and procedures as specified on drawing 9201268 or Government approved equivalent. The test shall be performed in accordance with Dwg. 9206748. A sample of two canister assemblies shall be selected in accordance with the sampling plan shown in 4.4.3.1. One canister assembly shall be tested for static luminosity and burning time. If the luminosity is less than 525,000 candlepower, the second candle shall also be tested. If the resulting average of the combined sample is less than 525,000 candlepower (luminosity), inspection lots represented by the samples shall be rejected. No one

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sample can exhibit a burning time of less than 60 seconds. In the event of failure of any sample to comply with the requirements as specified in paragraph 3.4.3, each individual inspection lot within the mix/batch must be tested. Failure of the inspection lot shall result in rejection of that particular inspection lot.

4.4.3.1 Sampling plan for luminosity and burning time. An inspection lot for the canister assembly shall be eighty-five candles produced consecutively from one mixing and pressing operation, and will be designated as one mix/batch for identification purposes (see 6.9). Smaller lot sizes will be accepted when dictated by interruption or shutdown of operations. The illuminant, as it is mixed, shall be dispensed into containers which shall be numbered and used in order. In the event of any production stoppage greater than two weeks, the sampling plan shall revert to level one, with only seven consecutive samples required to advance to each reduced level until the level of sampling in use when production was stopped is reached. If the mixing operation has not been shut down, but merely diverted to another item, and has not experienced any cause for rejection per the sampling plan of that item(s), the sampling level shall remain unchanged, and fifteen consecutive successful samples of two candles shall be required to advance to the next level, if applicable.

4.4.3.1.1 Level one sampling. A sample of two candles from each inspection lot shall be selected at random. The canister assembly shall be tested for compliance with the requirements specified in paragraph 3.4.3. This sampling frequency shall be continued until fifteen consecutive samples of two candles comply with the requirements of paragraph 3.4.3.

4.4.3.1.2 Level two sampling. After successful completion of level one sampling, the sampling frequency may be reduced to two samples from each group of six inspection lots. The canister assemblies shall be tested for compliance with the requirements specified in paragraph 3.4.3. In the event that one lot fails to comply with the requirements, the sampling shall revert to level one sampling.

4.4.3.1.3 Level three sampling. After successful completion of level two sampling, the sampling frequency may be reduced to two samples from each group of fifteen inspection lots. The canister assemblies shall be tested for compliance with the requirements specified in paragraph 3.4.3. This sampling frequency shall be continued until fifteen consecutive samples of two candles comply with the requirements of paragraph 3.4.3. In the event that one lot fails to comply with the requirements, the sampling shall revert to level two sampling.

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4.4.3.1.4 Level four sampling. After successful completion of level three sampling, the sampling frequency may be reduced to two samples from each shift (not to exceed ten hours) of production. The canister assemblies shall be tested for compliance with the requirements specified in paragraph 3.4.3. In the event that one lot fails to comply with the requirements, the sampling shall revert to level three sampling.

4.4.4 Air pressure test. The packed metal container assemblies shall be subjected to the internal air pressure test as specified on drawing 12934901. Observation shall be made for a pressure drop exceeding the requirement for the specified time. The cartridge packed in a defective container may be removed and packed in a replacement container to be tested.

4.4.5 Check test for possible deterioration of Primer, percussion, M28 (see MIL-P-46565 (AR)). If the total elapsed time between the original acceptance of the M28 primer lot and the assembly of that lot into the complete round cartridge exceeds two years, or if the M28 primers have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-P-46564 (AR) immediately before the M28 primer lot is assembled into the cartridges. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the complete cartridges (see 6.11).

4.4.6 Fit of projectile with cartridge case. The cartridge cases shall be gaged 100 percent for profile and diameter of the mouth to assure a proper fit with the projectile. Any case which fails to comply with the applicable requirements shall be classed defective and removed from the lot. (Note: this inspection shall not be required if it is determined that the cartridge cases have been previously gaged 100 percent for these characteristics).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The projectile metal parts contained in this specification are intended for use with Projectile, 105MM, Illuminating, M314A3, The M314A2 covered by this specification is military unique.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Requirements for submission for first article sample.

c. Requirements for submittal of Acceptance Inspection Equipment (see 6.3).

d. Requirements for ammunition lot numbering.

e. Submission of ballistic test data and ammunition data cards (see 6.4).

f. Requirements for submission of test equipment and procedure for fiberboard liner moisture content (see 6.5)

6.3 Submission of contractor acceptance inspection equipment (AIE) designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-QEM-A, Picatinny, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Submission of ballistic test data and ammunition data cards. In addition to the normal distribution of records for items procured by the Department of the Army, ammunition data cards should be forwarded to Commanding Officer, U.S Army ARDEC, ATTN: AMSRD-AAR-AEM-C and AMSRD-AAR-QEM-A, Picatinny Arsenal, NJ 07806-5000.

6.5 Fiber board liner moisture content. Test equipment and operating procedures for determining the fiberboard liner moisture content should be submitted for Government review and approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-QEM-A, Picatinny, NJ 07806-5000.

6.6 Visual examination qualification. When compliance with the applicable requirements is in doubt as a result of

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visual examination, this characteristic may be measured or gaged to determine acceptability.

6.7 Static burning time. Minimum static burning time as referenced herein (see 3.8.4) is required, as a time base for the determination of luminosity and compliance does not in itself guarantee final ballistic minimum burning time requirements. Structural integrity of the candle under set back loading is of prime importance in meeting the ballistic burning time requirement and is attained by proper formulation, blending, consolidation and cure.

6.8 Mix/batch. A mix of candles is defined as that quantity of candles which has been loaded with one mix of illuminant or al1 increments. A batch is defined as a quality of candles produced by a mixing process intended to make t h e product essentially uniform.

6.9 Functioning. "Satisfactory functioning" is defined as functioning of the fuze, ejection of the parachute and illuminant assembly, and proper suspension and ignition of the illuminant.

6.10 Effective illumination. Effective illumination is defined as burning of the canister without exploding or breaking up while suspended by the parachute. The time recorded for "effective illumination" may be considered as the time from the appearance of first light from the candle to either (1) burnout of the candle while still in the air or (2) burning of the candle until ground impact. Burning of the canister after striking the ground should not be considered as "effective illumination". Any occurrences which detract from "effective illumination" as herein defined should be reported in the official firing record.

6.11 Cost of check test. The contracting officer should arrange for the contractor to be reimbursed for the expense incurred in the performance of the check tests. The test should be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridges, and should not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.12 Drawings. Drawings listed in section 2 of this specification under the heading U.S Army Armament Research Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankfort Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

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6.13 Automatic gaging. Facilities equipped with Government furnished automatic gaging capable of performing 100 percent inspection of any characteristic should utilize capability in accordance with contract.

6.14 Process control dimensions. The contractor should prepare a process control plan for the dimensions marked process control. The plan should be submitted to the technical agency for approval prior to use. Dimensions to be gaged for process control are to be assured to a level IV.

6.15 Inspection equipment. For inspection equipment required to perform the examinations and tests prescribed herein the contractor should submit for approval inspection equipment designs in accordance with the terms of the contract. See paragraph 6.3 herein.

6.16 Automatic inspection equipment. All automatic test and inspection equipment when utilized by the contractor should be subjected to periodic verification or calibration checks during production. The procedures should include as a minimum: frequency, method of verification/calibration. These procedures should be included in the operation/calibration procedures as required by the above paragraph.

6.17 Critical classification notes.

6.17.1 Primer above flush. A primer above flush may result in unintended initiation of the primer causing the cartridge to function in the bore of the weapon, likely resulting in death or serious injury to the troops.

6.17.2 Projectile safety. Any premature function or metal parts separation in bore of the gun or in flight is likely to cause injury or death to troops firing the howitzer.

6.18 Definitions.

Acceptance Inspection Equipment (AIE). Any measuring device that is traceable to the national or international standard used to assure conformance of material to the contract requirements.

6.19 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes

6.20 Subject term (keyword) listing.

Ammunition
Artillery
Illuminating
Military Specification

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Munition
Projectile, 105MM, Illuminating, M314A3, Loading, Assembling and Packing

Custodian:
Army – AR

Preparing activity:
Army – AR
Project (1315-2008-001)

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